

Structured Reinforcement Learning beyond combinatorial decision-making : a primal–dual actor–critic for Atari benchmarks.

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01 Problem & Motivation

Reinforcement Learning (RL) : Agent learns by trial & error in order to maximize long-term reward through interaction with environment successes.

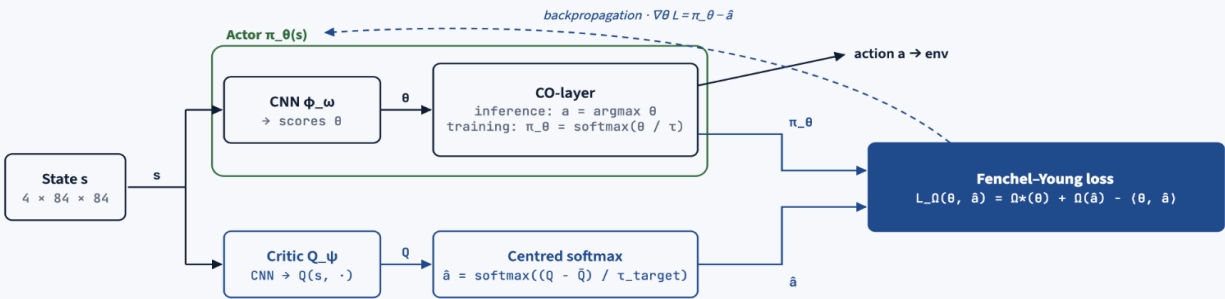
Structured RL (SRL) : introduced by Hoppe et al. (NeurIPS 2025), extends classical RL to stochastic combinatorial decision-making problems with a differentiable CO-layer enforcing feasibility on exponentially large action spaces : strong results on industrial / operational research problems.

Our question : Does the SRL framework still pay off outside of its original setting, on an other classic case of RL : Atari games?



02 Methodology – Adaptation of SRL

SRL framework is based on an actor-critic architecture with **Fenchel–Young loss** (smoothing the loss to integrate the non-differentiable CO-Layer), , we adapted the original pipeline. Objective: match the performance of (Mnih et al. 2013) DQN.



Stabilization tricks & Hyperparameters

State simplification : - reduction of pixels / frame / actions

Exploration strategy : - strong exploration at the beginning
-decreasing entropy reg during training

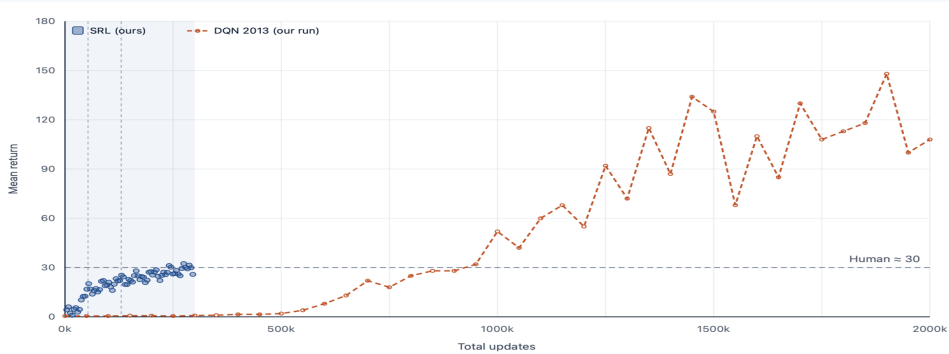
Training stabilization : - replay buffer to increase sample diversity
- hard update for the target network

Learning strategy : - increasing learning rate to follow a policy

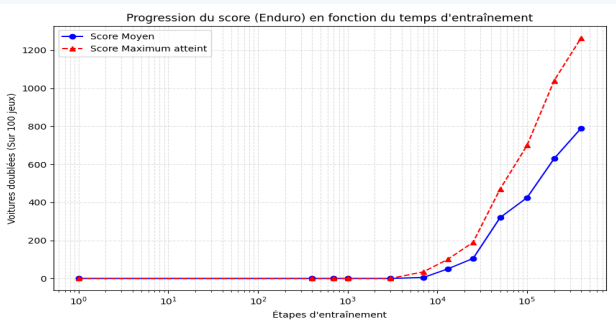
CNN of Breakout & enduro (same as DQN) : Conv(32, 8x8, /4) → Conv(64, 4x4, /2) → Conv(64, 3x3, /1) → FC(512, ReLU) → FC(|A|)

03 Experiments & Results

SRL VS. DQN 2013 — TOTAL UPDATES TO HUMAN-LEVEL on Breakout



SCORES on ENDURO



EXPERIMENTAL SETUP

- Breakout : Same hardware (Mac M3) for both runs, 50k update = 1h30 of runtime
- Enduro : Mac Mini M4, 50K updates = 1h20 of runtime

HEAD-TO-HEAD :

GAME	Enduro	Breakout	Pong	S. Invaders	Q*bert	Beam Rider
(MLP / CNN)	CNN	CNN	MLP	CNN	MLP	CNN
SIGNAL	Yes	Yes	No signal	No signal	Yes (poor)	Yes (Very poor)
BASELINE	DQN	DQN	Human level	Human level	Random	Human level

FINDINGS & LIMITS

Beat the DQN 2013 baseline where it could : SRL only worked on Breakout and Enduro, i.e. precisely the games that DeepMind had already beaten the human-level

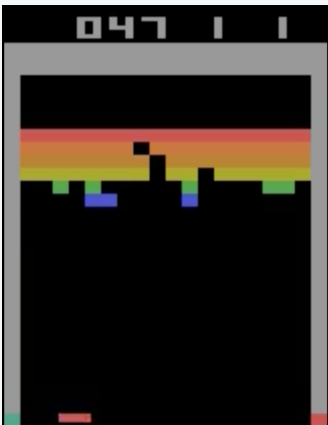
Off-label application: SRL is designed for large combinatorial spaces; on Atari we have 3-6 actions and our CO layer degenerates into a simple argmax. However, it still learns. Expand the action space by considering several consecutive actions did not bring improvement

Sparse reward = no policy: on Pong, Q*bert, etc., the agent fails to find exploitable trajectories → the method does not take off

Compute-bound: surpassing the ⅔-human plateau on Breakout or validating SRL on the other games would require much longer runs; no time no budget were available

Conclusion: SRL can generalize outside its original industrial settings even if its advantages appear stronger in larger combinatorial spaces

SRL AGENT · BREAKOUT



Frame from our best score on Breakout

VIDEOS

